

FINAL PRESENTATION SCRIPT

KineticPulse

A Multimodal Edge-AI Prototype for Fall Detection and Caregiver
Escalation

Embedded Systems, Cyber-Physical Systems and Robotics
Technical University of Munich

Speaker order: Youngwon Cho - Chang-Ting Zhong - Hao-Yuan Weng - Yule Xu
Ren-Yi Huang - Yuanhao Chen - Yiyuan Chen - Yong Zhe Sam

Presentation Order and Pacing

The allocation follows the slide order. The estimated total is about 15 minutes, including the demo video and short speaker handovers.

Presenter	Slides	Section	Target
Youngwon Cho	1-2	Opening and motivation	1:40
Chang-Ting Zhong	3-4	System journey and architecture	2:10
Hao-Yuan Weng	5-6	Prototype hardware and wearable link	1:55
Yule Xu	7	Visual pipeline	1:30
Ren-Yi Huang	8	Fusion and escalation	1:30
Yuanhao Chen	9	Incident-to-caregiver flow	1:20
Yiyuan Chen	10-11	Dashboard and live demo	2:40
Yong Zhe Sam	12-13	Evidence and conclusion	1:50

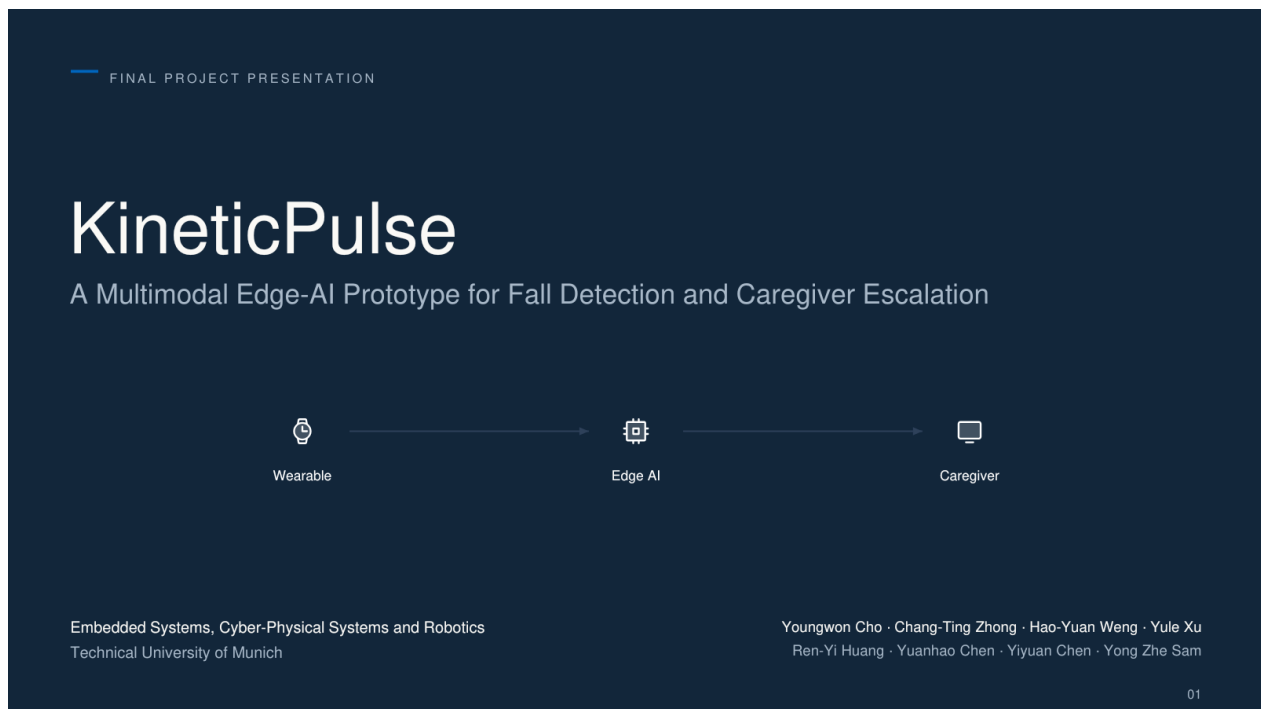
Delivery Notes

- Use the script as a guide rather than trying to imitate a memorized recording.
- Pause when changing sections, and point to the slide only when it helps the audience follow the explanation.
- NDJSON can be pronounced 'N-D JSON'; TSSTG can be pronounced as separate letters; mAP can be pronounced 'map'.
- The demo timing assumes a video of approximately 60 to 70 seconds.
- The italic delivery cues are not spoken aloud.

Slide 1 - KineticPulse

Presenter: Youngwon Cho

About 40 seconds



SCRIPT

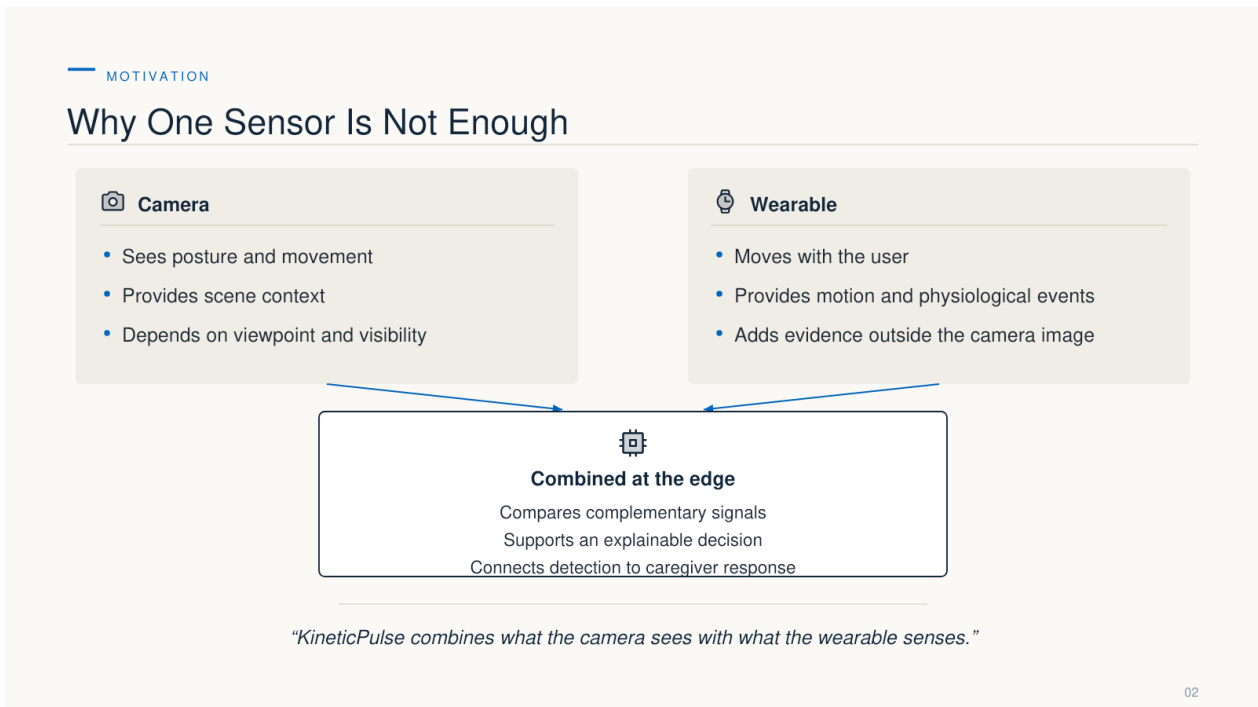
Hi everyone. We are the KineticPulse team, and today we are presenting our final project for Embedded Systems, Cyber-Physical Systems and Robotics. KineticPulse is an edge-AI prototype for fall detection. It combines a wearable, a camera, and a caregiver interface instead of depending on just one source of information. During the presentation, we will follow the same path as the system itself: from sensing, to understanding, to verification, and finally to the caregiver response.

Delivery cue: Open calmly. Look at the audience rather than reading the project title from the slide.

Slide 2 - Why One Sensor Is Not Enough

Presenter: Youngwon Cho

About 60 seconds



SCRIPT

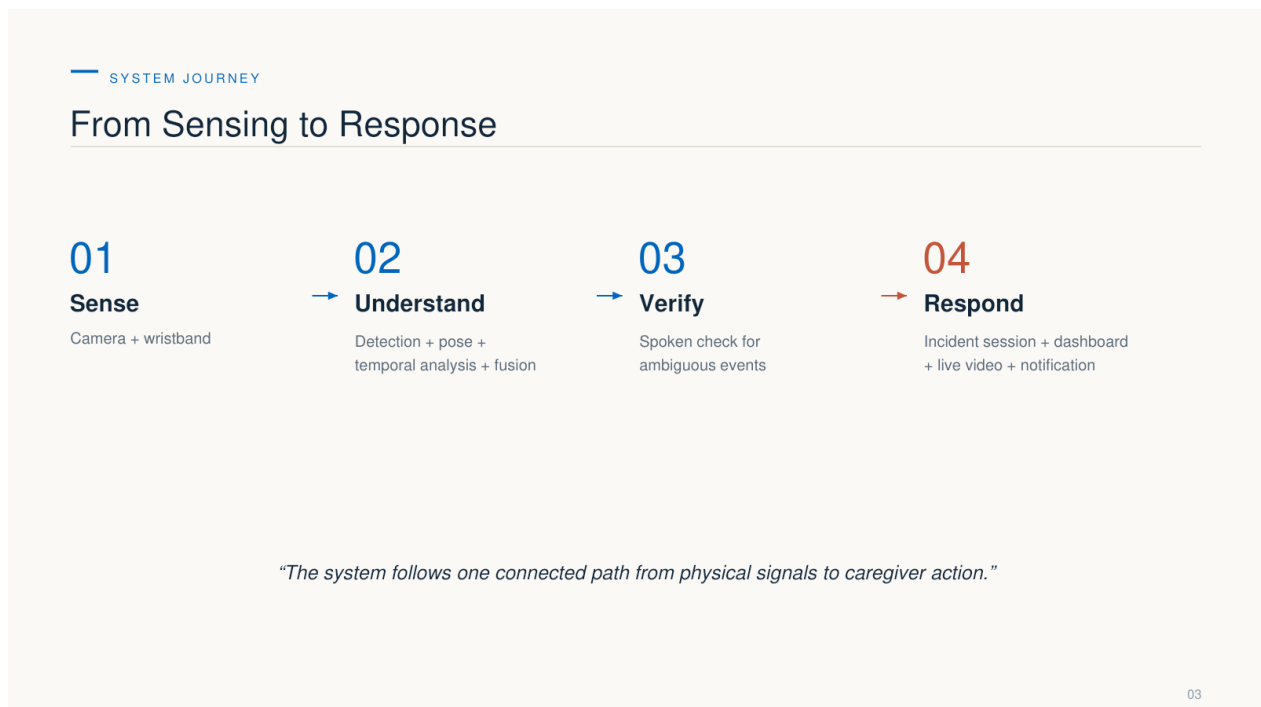
The main idea behind KineticPulse is that one sensor rarely tells the whole story. A camera is useful because it can see posture, movement, and the surrounding scene. But its view can be affected by the camera angle, lighting, or an object blocking the person. A wearable has different strengths. It stays with the user and can provide motion and physiological events even when the camera view is not ideal. We therefore bring both sources together on the Jetson. The system can compare what the camera sees with what the wearable senses before it decides how to react. This gives us a more understandable decision than simply trusting one signal. Chang-Ting will now show how this idea becomes one complete system flow.

Delivery cue: Point once to the camera column, once to the wearable column, and then to the combined box.

Slide 3 - From Sensing to Response

Presenter: Chang-Ting Zhong

About 55 seconds



SCRIPT

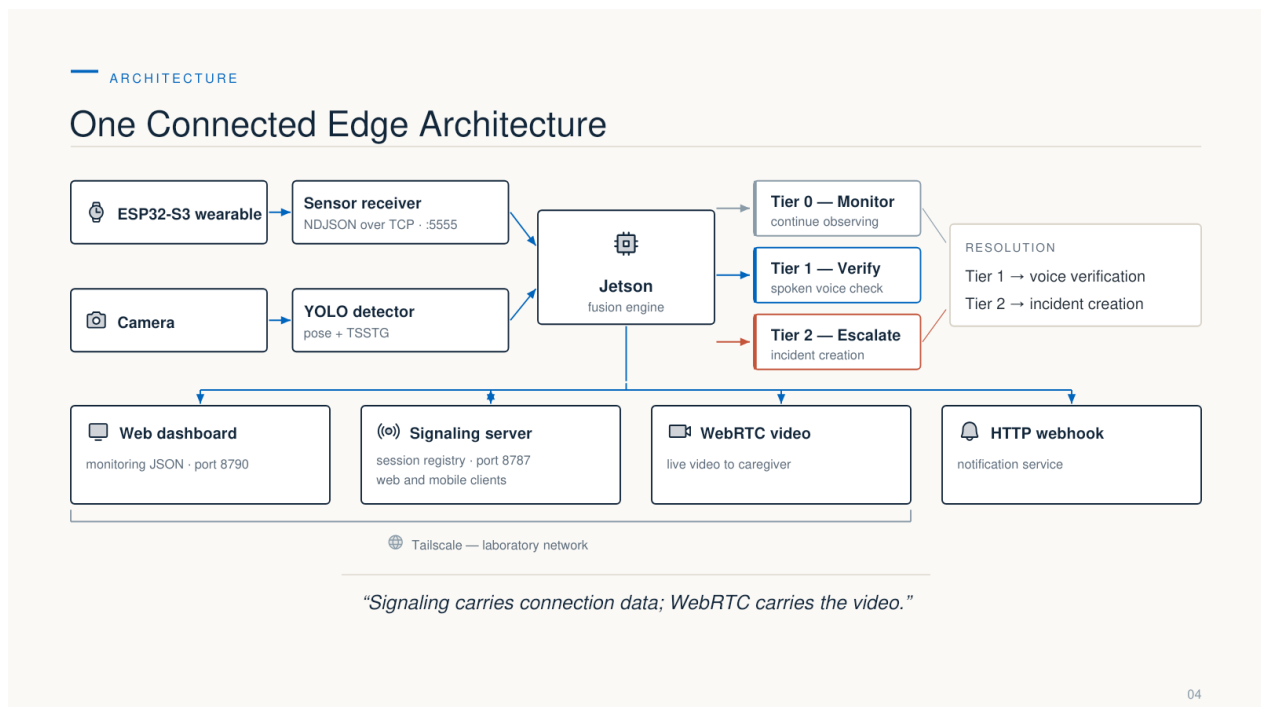
The easiest way to understand KineticPulse is through these four stages. First, we sense the situation using the camera and the wristband. Second, the Jetson tries to understand it. It runs posture detection, pose estimation, temporal analysis, and sensor fusion. Third, if the event is possible but not immediately critical, the system verifies it by asking the person whether they are okay. Finally, if help is needed, it responds by creating an incident session, updating the dashboard, sending a notification, and making live video available to the caregiver. So the project is not only about recognizing a fall. It covers the full path from the first signal to an actual caregiver view.

Delivery cue: Move through the four stages from left to right. Do not rush the final Respond stage.

Slide 4 - One Connected Edge Architecture

Presenter: Chang-Ting Zhong

About 75 seconds



SCRIPT

This diagram shows how the parts are connected. On the left, we have the two input paths. The ESP32-S3 wearable sends newline-delimited JSON to the Jetson over TCP port 5555. At the same time, the camera feeds the visual pipeline, which includes YOLO detection, pose estimation, and the TSSTG temporal model. Both paths meet in the fusion engine on the Jetson. The fusion result is one of three tiers: continue monitoring, verify by voice, or create an incident immediately. The lower part of the diagram shows the caregiver services. Port 8790 provides monitoring data for the dashboard, while port 8787 handles WebRTC signaling for the web and mobile clients. The video itself is carried separately through WebRTC, and a webhook provides the notification path. For our laboratory setup, Tailscale connects the Jetson and caregiver devices. Hao-Yuan will now take a closer look at the physical prototype and the wearable connection.

Delivery cue: Trace the diagram in this order: inputs, Jetson, tiers, then caregiver services.

Slide 5 - The Prototype We Built

Presenter: Hao-Yuan Weng

About 55 seconds

— PROTOTYPE

The Prototype We Built



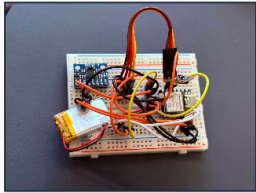
Assembled wearable enclosure with belt clip



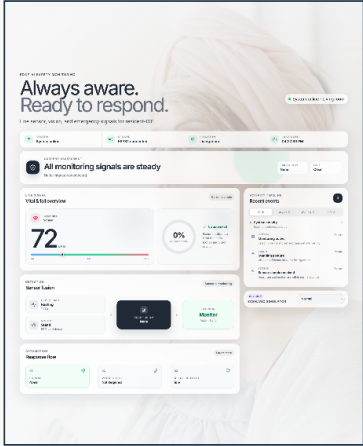
Opened enclosure with the prototype board inside



XIAO ESP32-S3, MPU-6050, MAX30102, and LiPo battery



Jetson Orin Nano in its 3D-printed case



KineticPulse web monitoring dashboard

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SCRIPT

Here are the main physical components and the web interface of our prototype. At the top left is the assembled enclosure with its belt clip. Next to it, you can see the enclosure opened with the prototype board inside. The lower hardware photos show the Jetson Orin Nano in its printed case and the breadboard with the XIAO ESP32-S3, the MPU-6050 motion sensor, the MAX30102 optical sensor, and the LiPo battery. On the right is the web monitoring dashboard. These parts represent the two sides of the project: the physical sensing and edge hardware, and the caregiver-facing software that turns the collected data into something useful.

Delivery cue: Briefly point to each photo in the same order as the script. Avoid describing small visual details.

Slide 6 - Wearable Data, One Event per Line

Presenter: Hao-Yuan Weng

About 60 seconds

WEARABLE LINK

Wearable Data, One Event per Line

ESP32-S3 wearable

CONNECTION

- Jetson listens on TCP port 5555
- ESP32 connects as the client
- Wi-Fi carries newline-delimited JSON

NDJSON OVER TCP

`{"type": "hello", ...}`

↓

`{"type": "hr", ...}`

↓

`{"type": "accel", ...}`

SUPPORTED EVENTS

hello — device and capabilities

hr — heart-rate value

accel — three-axis motion in g

ppg — infrared and red sample arrays

pulse_lost — loss-of-contact event

EXAMPLE

`</> {"type": "accel", "ax": 0.10, "ay": 0.02, "az": 0.99, "ts": 123456}`

"A simple typed contract keeps the wearable and Jetson independently testable."

06

SCRIPT

The wearable needs a simple and predictable way to send data to the Jetson. We use TCP, with the Jetson listening on port 5555 and the ESP32 connecting as the client. The data format is NDJSON, which means that every line contains one JSON event. We support five event types. Hello identifies the device and its capabilities. HR carries a heart-rate value. Accel carries the three motion axes in units of g. PPG contains infrared and red sample arrays, and pulse_lost reports a contact-loss event. The example at the bottom is one acceleration message with its three values and timestamp. Keeping every event small and typed makes the link easier to inspect, test, and extend. Next, Yule will explain what happens on the camera side.

Delivery cue: Pronounce NDJSON as 'N-D JSON'. Read the event names, but do not read the whole example character by character.

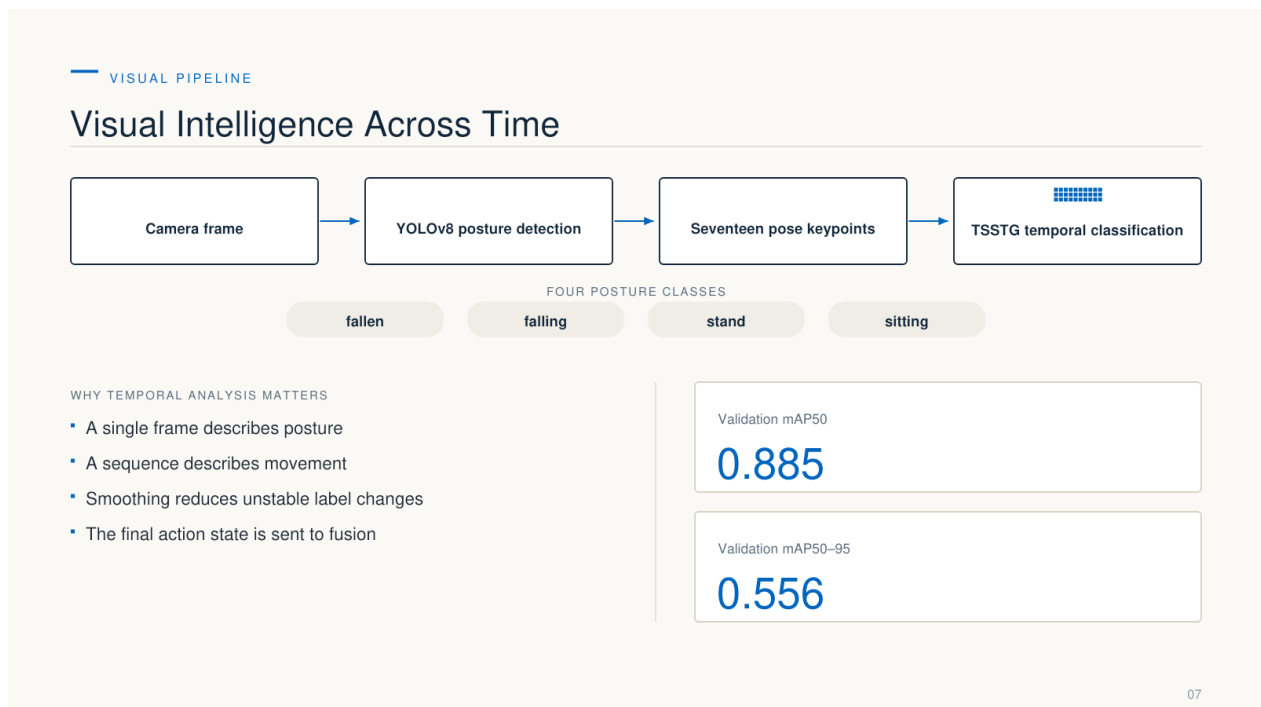
KineticPulse - Final Presentation Script

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Slide 7 - Visual Intelligence Across Time

Presenter: Yule Xu

About 90 seconds



SCRIPT

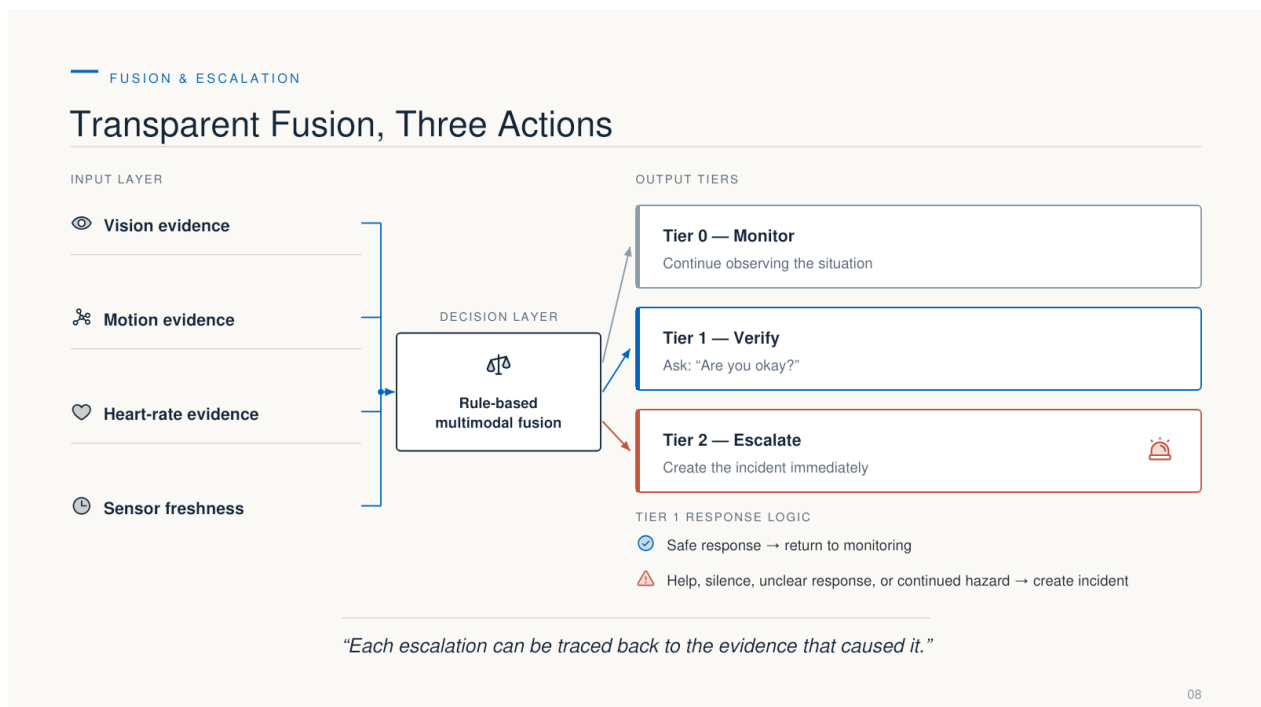
The visual pipeline starts with a camera frame. YOLOv8 first detects the person's posture, and the pose model extracts seventeen body keypoints. Those keypoints are then collected over time and passed to the TSSTG temporal classifier. This last step matters because a single image only shows a posture, while a sequence can show how the person moved into that posture. The pipeline uses four classes: fallen, falling, standing, and sitting. We also smooth the temporal output so that a short change in one frame does not immediately switch the final state. On the right are the reference validation results. The detector reached an mAP50 of 0.885. At the stricter mAP50 to 95 measure, the result was 0.556. In simple terms, the first metric evaluates detections at a 50 percent overlap threshold, while the second averages performance across increasingly strict thresholds. The stabilized action state is then passed to the fusion engine, which Ren-Yi will explain next.

Delivery cue: Pronounce TSSTG as separate letters. Pause briefly before explaining the two validation metrics.

Slide 8 - Transparent Fusion, Three Actions

Presenter: Ren-Yi Huang

About 90 seconds



SCRIPT

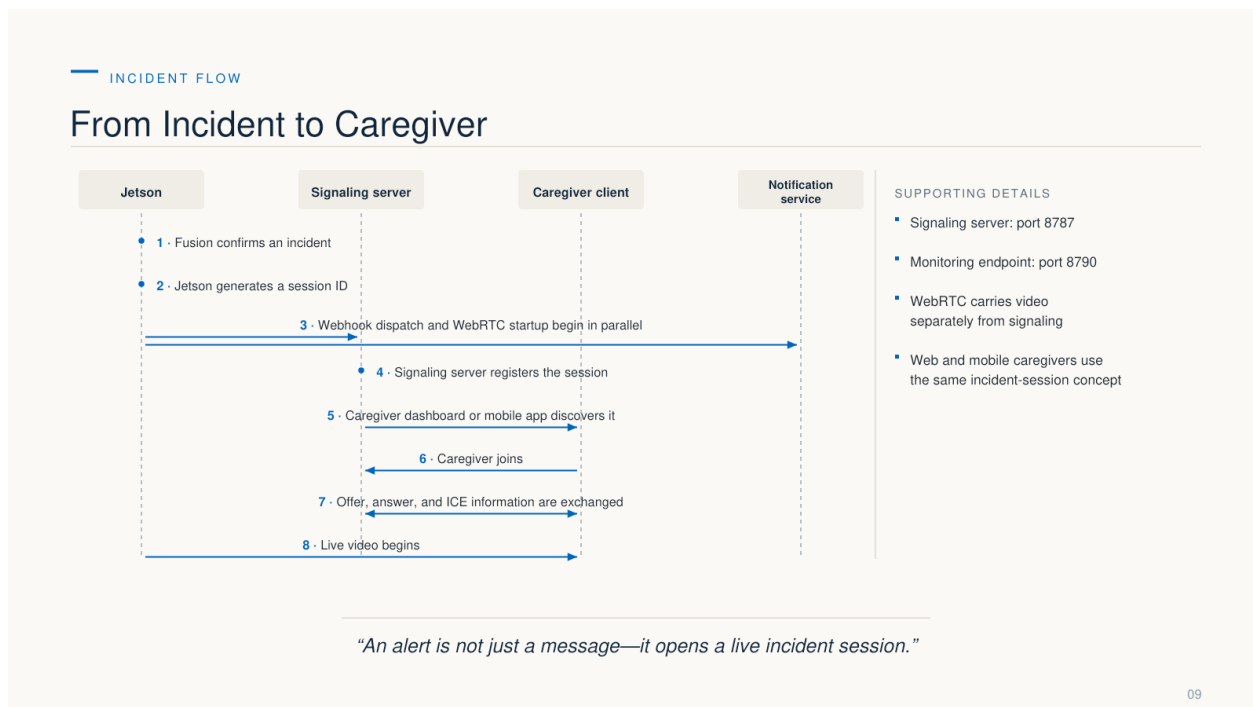
Fusion is the point where the visual and wearable information becomes one decision. The engine looks at four main inputs: the visual evidence, motion evidence, heart-rate evidence, and how recent the sensor data is. We chose explicit rules for this stage so that the reason behind each decision remains clear. There are three possible outputs. Tier 0 means that the system continues monitoring. Tier 1 means that the event should be checked, so the system asks the person, 'Are you okay?' If the person gives a clear safe response, the system returns to monitoring. If they ask for help, do not respond, give an unclear answer, or the hazard continues, an incident is created. Tier 2 is used when the combined evidence requires an immediate response, so the system creates the incident without waiting for the voice check. The important point is that each tier can be traced back to the evidence that produced it. Yuanhao will now show what happens after an incident is created.

Delivery cue: Use the three coloured tier boxes as anchors. Give Tier 1 slightly more time because it has two outcomes.

Slide 9 - From Incident to Caregiver

Presenter: Yuanhao Chen

About 80 seconds



SCRIPT

Once fusion confirms an incident, the Jetson creates a unique session ID. It then starts two paths in parallel. One path sends the webhook notification, while the other starts the WebRTC connection. The signaling server registers the new session, and the caregiver can discover it through either the dashboard or the mobile app. When the caregiver joins, the two sides exchange the WebRTC offer, answer, and ICE information needed to establish the connection. After that, the live video can begin. It is useful to separate these responsibilities: port 8787 handles the signaling and session information, while port 8790 provides the continuous monitoring data. The video stream is not sent through the signaling server. WebRTC carries it separately. So the caregiver receives more than a simple alert message. They receive a live incident session with the context needed to respond. Yiyuan will now show how this appears on the caregiver side.

Delivery cue: Follow the numbered sequence from left to right. Stress the difference between signaling and video.

Slide 10 - A Dashboard Built Around Decisions

Presenter: Yiyuan Chen

About 70 seconds

CAREGIVER DASHBOARD

A Dashboard Built Around Decisions

One normalized MonitoringModel → Overview Vitals Detection Events Live Sessions

WHAT THE CAREGIVER SEES

- Edge and wearable connection status
- Heart rate and fall confidence
- Vision and wearable inputs

TECHNICAL FOUNDATION

- GET /monitoring on port 8790
- Browser refresh approximately every three seconds
- Current fusion decision
- Voice and alert status
- Recent operational events
- SQLite event history through EventStore
- Monitoring and WebRTC signaling remain separate

"The interface turns sensor outputs into a clear response workflow."

real UI shown on Slide 5

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SCRIPT

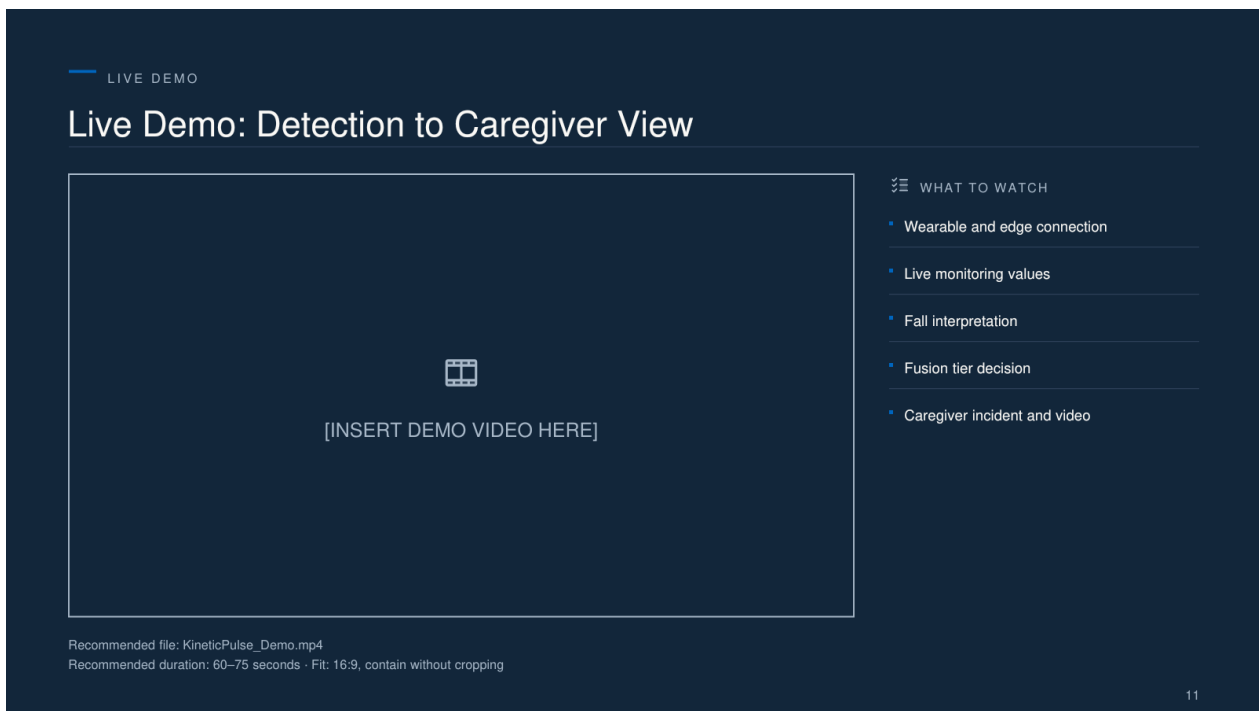
The dashboard brings the system information together in one caregiver view. All of its sections use the same normalized MonitoringModel, which keeps the data consistent across the Overview, Vitals, Detection, Events, and Live Sessions pages. The caregiver can see whether the Jetson and wearable are connected, the current heart-rate and fall-confidence values, the vision and wearable inputs, and the latest fusion decision. The interface also shows the voice-check and alert status, together with recent operational events. Technically, the dashboard reads the monitoring endpoint on port 8790 about every three seconds. Event history is stored through an EventStore interface backed by SQLite. We also keep monitoring separate from WebRTC signaling, so each path can be inspected independently. Now we will show the complete flow in the demo.

Delivery cue: Point to the five dashboard sections first, then move to the two bullet columns.

Slide 11 - Live Demo: Detection to Caregiver View

Presenter: Yiyuan Chen

About 90 seconds including the video

The slide has a dark blue background. At the top left, it says 'LIVE DEMO' in small white letters. Below that is the title 'Live Demo: Detection to Caregiver View' in a larger white font. In the center is a large white rectangular box containing a small video player icon and the text '[INSERT DEMO VIDEO HERE]'. To the right of this box is a sidebar titled 'WHAT TO WATCH' with a list of five items: 'Wearable and edge connection', 'Live monitoring values', 'Fall interpretation', 'Fusion tier decision', and 'Caregiver incident and video'. At the bottom left of the slide, there is small white text: 'Recommended file: KineticPulse_Demo.mp4' and 'Recommended duration: 60-75 seconds · Fit: 16:9, contain without cropping'. The number '11' is in the bottom right corner.

SCRIPT

Now we would like to show the system working from the edge device to the caregiver view. As the video plays, please watch the connection status, the live monitoring values, the fall interpretation, and the fusion decision. At the end, you will see the caregiver incident and the live-video step.

After the video: This short demo brings the separate parts together in the same order we have described: sensing, interpretation, fusion, and caregiver response. Finally, Yong Zhe will summarize the evidence across the whole pipeline.

Delivery cue: Start the video after the first paragraph. Pause for the video. Say the second paragraph only after it ends.

Slide 12 - Evidence Across the Whole Pipeline

Presenter: Yong Zhe Sam

About 75 seconds

EVIDENCE

Evidence Across the Whole Pipeline

 Perception	 Wearable communication	 Decision logic	 Caregiver path
- Four-class posture detector - Pose and temporal action processing - Validation mAP50: 0.885 - Validation mAP50-95: 0.556	- Typed NDJSON contract - TCP framing and reconnection handling - Motion, heart-rate, PPG, and contact events	- Fusion scenarios and escalation tiers - Temporal stabilization - Sensor-availability-aware decisions	- Monitoring publisher and dashboard - Incident sessions and signaling - WebRTC and webhook control flow

"The result is an integrated path from sensing to response—not a collection of isolated modules."

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SCRIPT

Before we finish, this slide summarizes the evidence for each part of the system. On the perception side, we have the four-class detector, pose estimation, temporal processing, and the validation results shown earlier. For wearable communication, we have the typed NDJSON contract, TCP framing and reconnection handling, and event types for motion, heart rate, PPG, and contact status. For the decision logic, we tested the fusion scenarios, escalation tiers, temporal stabilization, and behaviour when sensor availability changes. On the caregiver side, the system connects the monitoring publisher and dashboard with incident sessions, signaling, WebRTC, and the webhook path. The main result is therefore the integration of these parts. The project does not stop at a detector output; it connects sensing to a decision and then to a caregiver response.

Delivery cue: Move through the four columns at an even pace. Do not read every bullet word for word.

Slide 13 - Conclusion

Presenter: Yong Zhe Sam

About 35 seconds

We built one connected workflow that turns camera and wearable signals into an understandable caregiver response.

Thank you — Questions?

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SCRIPT

To conclude, KineticPulse combines camera and wearable signals on one edge platform. The Jetson interprets the data, applies an explainable escalation process, and connects the result to a caregiver through monitoring, notification, and live video. In short, we built one connected workflow from sensing to response. Thank you for listening. We are happy to take your questions.

Delivery cue: Slow down for the final two sentences and look at the audience before inviting questions.